MIL-B-80223
9 July 1971
SUPERSEDING
MIL-B-24048
17 March 1964

# MILITARY SPECIFICATION

BENDING MACHINES, PIPE AND CONDUIT, HYDRAULIC AND MECHANICAL, PORTABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE.

1.1 Scope. This specification covers hand operated-and electric motor driven, portable, hydraulic and mechanical, bending machines for bending pipe and rigid conduit.

1.2 Classification. Bending machines covered by this specification shall be of the following types, classes, and sizes, as specified (see 6.2.1):

Type I - Hydraulic

Class A - Electric Motor Driven

Size 1 - 1/2" thru 2" Standard Pipe and Rigid Conduit

Size 2 - 1-1/4" thru 4" Standard Pipe and Rigid Conduit

Size 3 - 2" thru 6" Standard Pipe and Rigid Conduit

Class B - Hand Operated

Size 1 - 1/2" thru 2" Standard Pipe and Rigid Conduit

Size 2 - 1-1/4" thru 4" Standard Pipe and Rigid Conduit

Size 3 - 2" thru 6" Standard Pipe and Rigid Conduit

FSC 3441

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Type II - Mechanical

Class A - Electric Motor Driven

Size 1 - 1/2" thru 2" Standard Pipe and Rigid Conduit

Class B - Hand Operated

size 1 - 1/2" thru 2" Standard Pipe

2. APPLICABLE DOCUMETNS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal fom a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-M-18058 | Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of. |
| MIL-P-15137 | Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-130 | Identification Marking of U. S. Military Property. |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

| | |
|---|---|
| Handbook H28 | Screw-Thread Standards for Federal Services. |

(Application for copiess should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS

| | |
|---|---|
| MG-1 | Motors and Generators. |
| IC- 1 | Standards for Industrial Controls. |

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

| | |
|---|---|
| 360.01 | Manual for Machine Tool Gearing. |

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, N. W., Washington, D. C. 20005.)

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the supplier shall furnish one complete machine of each type, class and size specified in the contract or order for first article inspection and approval in accordance with Section,4 herein.

3.2 Desire. The design shall enable the machine to make 90° bends in pipe and conduit using a hydraulic ram and bending form, a mechanically driven bending form, or a combination hydraulic and mechanical



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driven bending form without repositioning the workpiece. The machine shall be portable and consist basically of a frame, shoe, hydraulic pump and ram or a gear driven spindle. The machines furnished under this specification shall be new and of the manufacturer's current design. The design shall be such as to provide convenient, efficient, and safe operation. All parts subject to wear shall be easily accessible for adjustment, repair and replacement.

3.2.1 Safety devices. The machine shall be fitted with safety devices or coverings for all parts that present a safety hazard. The devices shall include guards for belts, pulleys, chains, sprockets and gears. All guards shalli provide easy access to guarded parts. When special safety devices are required, they shall be as specified (see 6.2.1).

3.2.2 Lubrication. The machine shall be provided with a means of lubricating all parts requiring lubrication. All lubrication devices shall be easily accessible. The machine shall be lubricated prior to delivery.

3.2.3 Interchangeability. All parts shall be manufactured to such standards as will permit adjustment or replacement without modification to parts or the machine.

3.2.4 Material. All material used in the manufacture of the machine shall be new and of the quality necessary to produce a machine to meet the requirements described herein.

3.3 Construction. The machine shall be constructed so that, when connected to the specified power source, when applicable, it can be used for any purpose for which it was designed. The machine shall be constructed from parts which are sound in structure and free from any repair.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall processes such as welding, peening, plugging, or filling with solders or metallic pastes be used on castings or forgings to reclaim any defective parts fox use on the machine.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are required in the



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original design. These operations shall not be employed as a repair measure for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners shall be installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free from fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. The condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

3.3.5 Painting. Unlees otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3.7 Gears. All gears shall be in the inch system and shall meet the applicable requirements of AGMA 360.01.

3.4 Performance. The bending machines covered by this specification shall produce 90° wrinkle-free bends with a minimum centerline radius not to exceed those allowed in Table I for the size machine specified. Smaller radii are acceptable.

3.5 Components.

3.5.1 Frame. The frame shall be a steel weldment, a steel casting, or a precision drilled and heat treated aluminum plate dependent upon the design of the machine. In all cases, the frame shall be adequate to support the components of the machine and shall transmit the bending force to the workpiece without permanent deformation.

3.5.2 Hydraulic system. The hydraulic system shall consist basically of a hydraulic pump, ram, reservoir, control, and hose or piping.



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TABLE I

| Rigid Conduit or Standard Pipe, Size (Inches) | Centerline Radius - Inches | | |
|---|---|---|---|
| | Size 1 | Size 2 | Size 3 |
| 1/2 | 4-1/2 | | |
| 3/4 | 6 | | |
| 1 | 7 | | |
| 1-1/4 | 9 | 9 | |
| 1-1/2 | 11 | 11 | |
| 2 | 13-1/2 | 13-1/2 | 13-1/2 |
| 2-1/2 | | 17-1/2 | 17-1/2 |
| 3 | | 20 | 20 |
| 3-1/2 | | 22-1/2 | 22-1/2 |
| 4 | | 26-1/2 | 26-1/2 |
| 5 | | | 34 |
| 6 | | | 54 |

3.5.2.1 Pump. The pump shall be hand operated or electric motor driven, and be capable of furnishing the required hydraulic power to bend workpieces of the maximum capacity of the machine,

3.5.2.2 Ram. The ram shall be designed to hold the bending shoes and shall provide the alignment necessary to obtain the accuracies specified herein.

3.5.2.3 Hose or piping. All hose or piping delivering fluid under pressure shall be capable of withstanding a hydrostatic test pressure of 8500 PSI. All hose shall be of ply construction with the inner ply of oil-proof synthetic rubber.



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3.5.3 Spindle. The spindle, or king pin, shall be of steel, designed to mount the bending shoes or radius blocks, and of sufficient strength to withstand the bending forces imposed.

3.5.4 Drive. The drive shall transmit power through a gear reduction unit, a gear train, or a chain and sprocket to the spindle or king pin. The power to the drive may be by an electric motor, a gear and ratchet mechanism, or any other suitable means.

3.5.5 Miscellaneous hardware. Each bending machine shall be furnished with all necessary miscellaneous hardware such as pins, clips, straps, pipe supports, etc. required by the design of the machine.

3.6 Size and characteristics. Unless otherwise specified (see 6.2.1), bending machines furnished under this specification shall be not less than as specified in Table II for the type, class, and size being procured.

3.7 Electrical equipment. Unless otherwise specified (see 6.2.1), all electrical components including motors, relays, switches, and wiring shall conform to and shall be located in accordance with NEMA Standards. The machine shall be completely wired through the controls and ready for connection to the applicable power source.

3.7.1 Motors. Motors shall be continuous duty and shall have ball bearings. Unless otherwise specified (see 6.2.1), motors shall be dual voltage 115/230 volt, single phase, 60 Hz, AC and initally wired for the operation on 115 volts. Unless otherwise specified (see 6.2.1), motors shall have a dripproof enclosure.

3.8 Equipment.

3.8.1 Standard equipment. Unless otherwise specified (see 6.2.1), the machine shall be furnished with all standard equipment and accessories normally furnished by the manufacturer.



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TABLE II

| Characteristics | Size 1 | Size 2 | Size 3 |
|---|---|---|---|
| | Type I, Class A | | |
| Workpiece Diameters<br>Workpiece Material | 1/2" thru 2"<br>Standard Pipe and Rigid Conduit | 1-1/4" thru 4"<br>Standard Pipe and Rigid Conduit | 2" thru 6"<br>Standard Pipe and Rigid Conduit |
| | Type I, Class B | | |
| Workpiece Diameters<br>Workpiece Material | 1/2" thru 2"<br>Standard Pipe and Rigid Conduit | 1-1/4" thru 4"<br>Standard Pipe and Rigid Conduit | 2" thru 6"<br>Standard Pipe and Rigid Conduit |
| | Type II, Class A | | |
| Workpiece Diameters<br>Workpiece Material | 1/2" thru 2"<br>Standard Pipe and Rigid Conduit | | |
| | Type II, Class B | | |
| Workpiece Diameters<br>Workpiece Material | 1/2" thru 2"<br>Standard Pipe | | |



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3.8.2 Optional equipment. When required (see 6.2.1), the following optional equipment shall be furnished as specified.

a. Bending shoes (extended capacity or reverse bending type).

b. Bending shoes (standard shoes when not furnished with the machine).

c. Shot gun type shoes.

d. Radius bending block. -

e. Slide blocks.

f. Mandrels.

g. Metal storage box with lid, hinges, and clasp.

3.8.3 Repair Parts. Such repair parts as are specified (ace 6.2.1), shall be provided.

3.8.3.1 Provisioning (For Naval Shipboard Application). Provisioning technical documentation shall be in accordance with method B of MIL-P-15137. Determination for quantity of repair parts shall be at the discretion of the Program Support Inventory Control Point, Ships Parts Control Center, Mechanicsburg, Pennsylvania 17055.

3.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, marked in accordance with MIL-STD-130, shall be securely affixed to each machine, The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number
power input characteristics and ratings
Date of manufacture
Contract or order number
Federal Stock Number or Plant Equipment Code
U S



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3.10 Identification plate. When specified (see 6.2.1), a separate identification plate of corrosion resistant metal permanently and legibly marked in letters 3/16 inch high with the caption "Departmental or Activity Identification" shall be securely affixed to each machine. The plate shall have blank space for the Government to stamp 25 letters or numbers of the same size as the caption.

3.11 Workmanship. Workmanship of the machine and accessories shall be commensurate with the requirements specified herein and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article sample when required in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examinaticn in 4.5 and all tests in 4.6, Failure of the first article to pass the examination or any first article test shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each bending machine being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality



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conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the machine to pass the examination, test, or inspection shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Operational test. Hand operated machines shall be operated through a complete cycle to determine the bender works properly. Electric motor driven machines shall be operated for a minimum of five minutes during which time all components shall be observed for proper operation. Each bending shoe, radius block, slide block, or mandrel furnished with the machine shall be mounted to the machine and checked for proper fit.

4.6.2 Performance test. The machine shall be fitted in turn with the minimum and maximum size bending forms furnished with the machine. One test bend each will be produced, utilizing pipe or conduit, of the size applicable for the size machine under test. Scrap pipe or conduit may be used, provided it will develop the bending capacity of the machine. Failure of the machine to produce hands in accordance with the requirements of Table I shall be cause for rejection.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable requirements of MIL-M-18058. Level of preservation, packaging, and packing shall be as specified (see 6.2.1).



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6. NOTES

6.1 Intended use. Bending machines covered by this specification are intended for use in maintenance, repair, and rebuild facilities for bending pipe, tubing, and conduit.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

l . Title, number, and date of this specification.

b. Type, class, and size required (see 1.2 and 3.6).

c. First article, if required (see 3.1).

d. Special safety devices, if required (see 3.2.1).

e. Painting, if different (see 3.3.5).

f. Sizes and characteristics, if different (see 3.6).

g. Electrical equipment, if different (see 3.7).

h. Motor and voltage characteristics, if different (see 3.7.1).

i. Voltage of initial wiring, if different (see 3,7.1).

j. Motor enclosure, if different (see 3.7.1).

k. Standard equipment, if different (see 3.8.1).

1. Optional equipment, if required (see 3.8.2).

m. List of repair parts required (see 3.8.3).

n. Nameplate, if different (see 3.9).

o. Identification plate, when required (see 3.10).



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p. First article inspection, if different (see 4.3).

q. Quality conformance inspection, if different (see 4.4).

r. Selection of level of preservation, packaging, and packing (see 5.1).

6.2.2 Contract data requirements. Technical data such as the manufacturer's standard operators manual, parts list, foundation and anchor bolt plans, and wiring diagrams will usually be required for delivery in connection with this specification. When technical data is required, it will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 Incorporated into the contract.

6.3 Instructions for first article approval. When first article inspection and approval is required for equipment furnished under this specification, the solicitation should conform to the applicable requirements of Section I, Part 19 of the Armed Services Procurement Regulation.

6.4 Supersession data. The size 1, 2, and 3, hand operated, hydraulic bending machines previously covered by MIL-B-24048 are now covered by the Type I, Class B, size 1, 2, and 3 of this specification. All other types, classes, and sizes covered by this specification have not been previously covered.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH, OS
Air Force - 84

User Activities:
Navy - MC, AS

Preparing Activity:
DSA- IP

Project Number:
3441-0038



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# SPECIFICATION ANALYSIS SHEET

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government use of the specification in procurement of products for ultimate use by the Depar is provided for obtaining information on the use of this specification which will can be procured with a minimum amount of delay and at the least cost. Commen will be appreciated. Fold on lines on reverse side, staple in corner, and send to and suggestions submitted on this form do not constitute or imply authorization referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED IN MENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specificati attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH

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Commander
Defense Industrial Plant
ATTN: DIPEC-TF
Memphis, Tennessee 38102

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